Phongsakon Mark Konrad

Curriculum Vitae

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Google Scholar

Summary

Master of Science in Machine Learning student at the University of Tübingen supported by the Research-Oriented Master's Scholarship of the Zuse School ELIZA, with a background in empirical machine-learning research and research engineering. My research interests are in causality, in particular the foundations of causal learning: what allows a learning system to move from statistical prediction toward causal understanding. I use deep learning theory and the mechanics of learning, how optimization shapes learning dynamics and the representations neural networks learn, to study when and how such structure emerges. I am moving toward theoretical work while bringing empirical intuition, controlled experimentation, and software engineering experience.

Education

2026– **Master of Science in Machine Learning**, *University of Tübingen*, Tübingen, Germany
Research-oriented, four-semester programme; **Master's Scholar, Zuse School ELIZA** (Research-Oriented Master's Scholarship). Focus on probability, statistics, optimization, and learning theory.

2025 **Exchange, Computer Science & Engineering**, *Hong Kong University of Science and Technology*, Hong Kong SAR
Courses in machine learning, deep learning in computer vision, large language models, reinforcement learning, and postgraduate data visualisation.

2023–2026 **Bachelor of Science in Engineering, Software Engineering**, *University of Southern Denmark*, Sønderborg, Denmark
Graduated June 2026. Danish weighted average: **10.71/12** over 140 numerically graded ECTS (approximately **3.71/4.0**, WES-style U.S. equivalent). Co-authored the bachelor thesis with Tim Lukas Adam (equal contribution); **selected as the programme's best in a graduating cohort of more than 100 students.**

Research Experience

Oct 2026– **Research Collaborator**, *Applied and Interpretable Machine Learning Research Group, University of Southern Denmark*, Remote
Continue to conduct research with the group after completing the BSc, as a member of its research team with access to its compute resources.

Jan–Oct 2026 **Undergraduate Student Researcher**, *Applied and Interpretable Machine Learning Research Group, University of Southern Denmark (Supervisor: Serkan Ayvaz)*, Sønderborg, Denmark
Independently initiated and led research projects, owning problem formulation, experimental design, implementation, analysis, and manuscript preparation; first author on multiple resulting publications.

Dec 2025–Sep 2026 **Research Collaborator**, *Hong Kong University of Science and Technology DataVISards (Supervisor: Arpit Narechania)*, Hong Kong SAR
Contributed to a collaborative visualisation research project established during a 2025 exchange semester, resulting in a paper accepted at IEEE VIS 2026.

Sep 2024–Dec 2025 **Student Research Assistant**, *University of Southern Denmark Data & Intelligence Lab*, Sønderborg, Denmark
Built machine-learning pipelines and contributed to study design, literature review, manuscript preparation, grant proposals, and multimodal dataset management across lab projects, alongside my own first-author research.

Selected Publications

Chosen for fit with my research question: when what a model represents is also what it uses, and what evidence identifies the difference. Full list on my website and Google Scholar.

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Routing Subspaces: Auditing Evaluation-to-Deployment Mismatch in Fine-Tuned Language Models*. *Conference on Neural Information Processing Systems (NeurIPS 2026)*, accepted. [arXiv]
Localizes the internal coordinate separating evaluation from ordinary use after fine-tuning; intervening on it closes the gap in ten of twelve model–behavior settings.

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Self-Reports Do Not Identify Self-Models: An Identifiability Test for Counterfactual Reports*. Oral presentation, *Philosophy of Machine Learning Workshop at the International Conference on Machine Learning (ICML 2026)*. [OpenReview] [arXiv]
Environment-shift invariance tests across three open models separate report accuracy from mechanism binding.

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout*. Poster, *Workshop on Mechanistic Interpretability at the International Conference on Machine Learning (ICML 2026)*. [OpenReview]
Matched pretrained/instruction-tuned models with probes, activation patching, head ablations, subspace controls, and cross-model replication.

2026 **P. M. Konrad**, A.-A. Popa, Y. Sabzehmeidani, L. Zhong, M. Tripathy, A. Constantinescu, E. A. Liehn, S. Ayvaz. *Challenges in Deep Learning-Based Small Organ Segmentation: A Benchmarking Perspective for Medical Research with Limited Datasets*. *Biomedical Signal Processing and Control* 128, 111226 (Q1, SJR). [Published version] [arXiv]
Ten-model benchmark with ablations, bootstrap uncertainty, and independent distribution-shift evaluation for small cardiovascular histology datasets.

Teaching

Jan–Aug 2026 **Teaching Assistant, Artificial Intelligence**, *University of Southern Denmark*
Led exercise sessions, supported students with Artificial Intelligence and Machine Learning concepts, helped design hands-on lab assignments, and developed interactive supplementary course material.

Invited Talks

Feb 2026 **Agentic Engineering in Practice**, House of Software Sønderborg inaugural event; invited talk to 100+ attendees.

Academic Service

2025–2026 **Student Member**, Educational Committee, Software Engineering Programme, University of Southern Denmark. Represented students in curriculum and quality-assurance discussions concerning course content, assessment, and the study environment.

Selected Recognition

2026 **Best bachelor thesis**, Software Engineering programme, University of Southern Denmark; co-authored with Tim Lukas Adam (equal contribution).

2026 **Best Startup Award finalist** (one of five), DreamBear, SDU Startup Night.

2025 **Top 10**, Danish National Championship in Artificial Intelligence; individual competitor, organised by the Danish Society for Artificial Intelligence.

2025 **Travel grant**, Danish Data Science Academy (DDSA), to present my first paper at KES 2025.

2024 **Top 10**, Danish National Championship in Artificial Intelligence; team competitor, organised by the Danish Society for Artificial Intelligence.

Military Service

2017–2021 **Staff Duty Soldier**, *Bundeswehr, German Armed Forces*, Glücksburg, Germany
Led a small HR team administering more than 600 soldiers. Received a **Formal Recognition for Exemplary Service** and a **Performance Bonus for Outstanding Achievement** in 2021.

Additional Experience

Jan–Jun 2026 **Founder**, *SaturoLabs*, Denmark
Built and shipped DreamBear, an iOS app creating personalised narrated bedtime stories that frame neurodivergent traits as strengths.

Jun–Nov 2024 **CTO & Co-Founder**, *Tutora ApS*, Sønderborg, Denmark
Led end-to-end development of the company's web application.

2022–2024 **Co-CEO & Co-Founder**, *Yeager GmbH*, Remote
Built an AI-assisted powerlifting training application.

Technical Skills

Research: Experimental design, controlled interventions, activation patching, representation probing, ablations, bootstrap uncertainty, model training and evaluation, segmentation, conformal prediction, reproducibility.

Machine Learning: Python, PyTorch, Hugging Face Transformers, TransformerLens, scikit-learn, pandas, NumPy, Optuna, Weights & Biases.

Engineering: Git, Docker, Linux, SQL, Google Cloud Platform, JavaScript.

Languages

Fluent: English, German

Basic: Danish, Thai

Last updated 3 October 2026.